

INDUCED STRUCTURE — UPDATED RESEARCH PAPER

THE LIVING ENCYCLOPEDIA

Question-Induced Knowledge Architecture, Recursive Vault Enrichment,
and the Formation of Institutional Intelligence

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Research paper and computational experiment

Central proposition. A language model and a proprietary vault can jointly produce a changing epistemic environment. Purposeful questions induce candidate relations; environmental shocks revise them; governed write-back converts selected reasoning episodes into persistent external memory. The model's parameters need not change for the model–vault system to learn.

Companion computational laboratory. This paper is accompanied by the Google Colab notebook *MA_Induced_Context_Graph_Complete_Experiment.ipynb*. It creates a 100-company Obsidian vault, evaluates all 4,950 possible company pairs, records four question-driven reasoning rounds, injects macroeconomic and innovation shocks, generates typed graph links, and exports before-and-after visualizations, provenance records, checksums, and an audit bundle.

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1 Abstract

Retrieval-augmented generation is commonly described as a way of giving a language model access to information outside its parameters. That description is useful but incomplete for systems in which model outputs are selectively written back into a persistent knowledge environment. In such systems, the external corpus is no longer a passive source of context. It can become the record of prior analytical encounters: the questions that were asked, the relations proposed in answering them, the evidence used, the conditions under which a judgment held, and the events that later strengthened, weakened, or invalidated it.

This paper develops the concept of **Question-Induced Knowledge Architecture** (QIKA). The architecture begins with a proprietary vault of source-grounded facts. A consequential question gives analytical purpose to the interaction. A large language model combines retrieved proprietary evidence with its parametric priors and proposes new symbolic structure: typed relationships, hypotheses, scenario exposures, and recommendations. After validation and epistemic labeling, selected objects are written into a derived graph layer. Subsequent questions retrieve not only the original corpus but also the governed record of earlier reasoning. Context is therefore jointly produced by the vault and the model across time.

The resulting process is neither conventional training nor ordinary retrieval. The model weights may remain fixed, yet the larger cognitive system changes as its external memory accumulates reusable, question-conditioned experience. The paper distinguishes this mechanism from traditional RAG, GraphRAG, fine-tuning, self-refinement, and agent-memory architectures while recognizing its intellectual proximity to each. It then presents a computational experiment based on a synthetic 100-company M&A universe. The experiment moves from disconnected company facts to initial merger hypotheses, macro-shock vulnerability links, innovation-created opportunities, and final integrated recommendations. Every edge is attributable to a named question, reasoning episode, condition, and epistemic status.

The contribution is a theoretical proposition, a governed implementation pattern, and a falsifiable experimental design. The strongest claim is not that an LLM can add links to a graph. It is that purposeful inquiry can externalize model-mediated analytical experience into a persistent, inspectable structure that improves later decision work without confusing generated hypotheses with facts.

2 From Context Windows to Cumulative Knowledge Formation

2.1 The familiar picture: a model receives context

The standard picture of applied generative AI is episodic. A user presents a question. A system retrieves relevant documents. The model receives those documents in its context window and produces an answer. Once the answer is delivered, the underlying corpus remains substantially unchanged. The next question begins with the same documentary environment, except for whatever conversation history happens to remain available.

This picture has been enormously productive. It separates parametric memory from external information, permits access to private or recently updated documents, improves provenance, and avoids the expense of retraining a model whenever a corpus changes. The original RAG formulation explicitly combined a pretrained sequence model with a non-parametric document index [3]. Yet the ordinary implementation treats the corpus as a library from which passages are borrowed, not as a body of knowledge that is transformed by the act of inquiry.

The deeper possibility arises when the answer to one question changes what will be available to answer the next. Suppose an initial vault contains company profiles but no asserted relation among them. The question “Which companies are plausible merger candidates?” forces a comparison across capabilities, geographies, balance sheets, and growth profiles. The response may contain relations that were not written anywhere in the original corpus. If those relations are merely displayed, the interaction is ephemeral. If they are typed, justified, reviewed, and stored, the vault has acquired a new layer of analytical structure.

2.2 The encyclopedia that writes new volumes

An intuitive analogy is a living encyclopedia. The initial volumes contain source material. A reader asks a substantive question that requires synthesis across several entries. The resulting analysis is not simply another copy of the original pages. It is a new, conditional chapter: a proposed explanation, a comparison, a map of dependencies, or a decision hypothesis. If this new chapter is reviewed and added to the collection, later authors can build upon it. When the environment changes, a further volume records which conclusions survived, which failed, and why.

This analogy captures three important properties.

1. **The new material is jointly produced.** Proprietary evidence constrains the analysis; the model contributes general schemas, analogies, taxonomies, and reasoning patterns encoded in its parameters.
2. **The sequence matters.** A later analysis is conditioned on the results, uncertainties, and failures of earlier analyses. The system is path-dependent.
3. **The collection changes its own future use.** New nodes and edges alter retrieval paths, expose previously distant evidence, and make different multi-hop arguments available.

The important phrase is *jointly produced context*. The context at time $t + 1$ is not simply the original corpus plus more external documents. It is partly the preserved outcome of the system’s own earlier encounters with that corpus. Generative AI therefore becomes more than a mechanism for responding within context. It becomes a participant in the governed formation of future context.

3 The Central Thesis: Questions Induce Experience

Data does not create analytical experience by itself. A purposeful question establishes relevance, comparison, and a decision criterion. The reasoning episode that answers it creates candidate experience. Environmental shocks then revise that experience. Governed persistence turns the sequence into institutional memory.

3.1 Why the question is a causal object

A graph cannot be meaningfully enriched by asking a model to “add useful links” in the abstract. Usefulness is relative to an objective. Two companies may be related because they share a sector, compete for the same customers, depend on the same infrastructure, hedge one another’s risks, possess complementary technologies, or become attractive only under a particular regulatory event. A generic similarity relation collapses these distinct meanings.

A question supplies a temporary ontology of relevance. “Which firms could merge?” privileges compatibility and synergy. “Which firms are vulnerable to higher rates and currency depreciation?” privileges leverage, liquidity, interest sensitivity, and foreign-exchange exposure. “Which new combinations become valuable after an agentic-AI breakthrough?” privileges data, distribution, cloud, cybersecurity, automation, and integration readiness. Each question exposes a different relational surface in the same underlying company universe.

The question should therefore be represented as a first-class node rather than discarded after prompting. Its text, criteria, date, author, scope, and parent question are part of the provenance of every relation it induces. A link between Company A and Company B does not mean “A is related to B” in some universal sense. It means, for example:

Under question Q001, using evidence set S_t , model configuration M , and pre-shock assumptions C_t , the system proposed that A and B constitute a high-ranking M&A hypothesis, with score s , rationale r , and confidence c .

3.2 From response to reasoning episode

A response becomes induced experience only when it is transformed into an inspectable object. At minimum, a reasoning episode should preserve:

- the initiating question and decision criteria;
- the retrieved source evidence and its provenance;
- the candidate entities and relations considered;
- the selected output and relevant rejected alternatives;
- assumptions, conditions, uncertainty, and model configuration;
- the relation to earlier episodes; and

- the governance decision: rejected, quarantined, provisional, approved, superseded, or invalidated.

This is not equivalent to storing a conversation transcript. A transcript is chronological but weakly structured. Induced experience is deliberately symbolized so that later retrieval can address questions such as: Which recommendations depended on low interest rates? Which company pairs appeared only after an innovation shock? Which conclusion was invalidated, by what event, and on whose approval?

4 Formal Construction: A Fixed Model, a Learning System

Let the protected source corpus at time t be S_t . Let D_t denote a derived, governed knowledge layer containing typed graph objects. The usable vault is

$$V_t = S_t \cup D_t.$$

Let M denote the language model with fixed parameters θ , q_t the purposeful question, x_t any external event or shock, and $\mathcal{R}$ the retrieval operation. The model receives:

$$C_t = \mathcal{R}(q_t, V_t) \cup \{x_t\}.$$

The reasoning function generates a candidate knowledge increment:

$$\widehat{\Delta K}_t = F_\theta(q_t, C_t, \Pi_t),$$

where Π_t contains the prompting protocol, schema, decision criteria, and tool outputs. The candidate increment may include nodes, typed edges, rankings, counterarguments, and revision operations. It is not yet canonical knowledge.

A governance operator Γ tests provenance, epistemic class, consistency, confidence, policy compliance, and human approval:

$$\Delta K_t^{\text{approved}} = \Gamma(\widehat{\Delta K}_t; S_t, D_t).$$

The derived vault then evolves as:

$$D_{t+1} = D_t \Delta K_t^{\text{approved}} - \Delta K_t^{\text{superseded}}, \quad V_{t+1} = S_{t+1} \cup D_{t+1}.$$

The minus sign does not imply destructive deletion. In a well-governed implementation, superseded objects remain historically available but are removed from the set of currently valid claims. Their temporal status and replacement lineage are preserved.

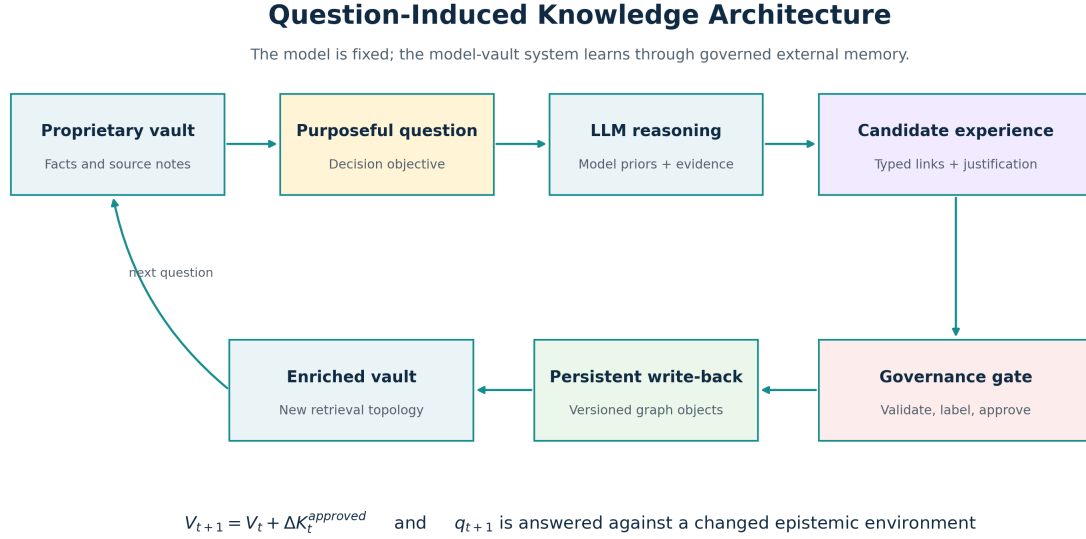


Figure 1: Question-Induced Knowledge Architecture. Questions convert retrieved facts and model priors into candidate experience. Governance and persistent write-back change the knowledge environment encountered by the next question.

4.1 System learning without parameter learning

No gradient update is required in this construction:

$$\theta_{t+1} = \theta_t.$$

Nevertheless, system performance may change:

$$P(M, V_{t+1}, q) \neq P(M, V_t, q),$$

because V_{t+1} contains validated relational structure that was absent from V_t . The system has learned in a non-parametric, externalized sense. This is closer to the accumulation of a research archive, case-law tradition, investment-committee memory, or scientific notebook than to conventional model training.

The distinction is essential. Saying that “the LLM learned” would be misleading if its weights did not change. Saying that “nothing learned” would also be misleading if the combined system can now retrieve prior hypotheses, trace their conditions, compare them with later shocks, and perform better on held-out decisions. The correct unit of analysis is the model–retriever–vault–governance system.

4.2 The context-compounding hypothesis

If approved structure is useful, a sequence of inquiries may exhibit contextual compounding. Define performance on a held-out task distribution $\mathcal{Q}^*$ as:

$$\mathcal{P}_t = \mathbb{E}_{q \sim \mathcal{Q}^*} [U(F_\theta(q, \mathcal{R}(q, V_t)))] ,$$

where U combines factual accuracy, decision utility, calibration, and provenance quality. Contextual compounding exists when:

$$\mathcal{P}_{t+1} - \mathcal{P}_t > 0$$

for a meaningful range of t , after controlling for additional source documents and token budget. The hypothesis is not that performance increases forever. Saturation, redundancy, and error accumulation imply a likely concave trajectory. The empirical question is whether governed, non-redundant experience produces a measurable positive region before diminishing returns or epistemic pollution dominate.

5 What Exactly Is Being Projected?

It is tempting to say that the model projects its “wisdom” into the vault. That phrase is evocative but scientifically imprecise. A pretrained model contains a compressed statistical representation of patterns in its training data. It can supply broad conceptual priors: common causal mechanisms, industry structures, financial relationships, taxonomies, analogies, and forms of argument. These priors are powerful, but they are neither guaranteed facts nor institution-specific knowledge.

The vault supplies what the model lacks: proprietary observations, current states, organizational definitions, transaction history, constraints, decisions, and local meaning. The induced relation is produced by binding general model priors to specific vault evidence under a purposeful question. We may call this mechanism **latent-to-symbolic projection**, provided that three qualifications remain explicit:

1. the latent representation is not directly observed;
2. the symbolic output is a model-mediated proposal, not a recovered truth; and
3. validation is required before the output changes the institution’s trusted knowledge state.

The most valuable output may be a relation that is not a close embedding neighbor. Similarity retrieval can readily connect two notes using the same vocabulary. More interesting synthesis occurs when the model links distant objects through a defensible mechanism: a logistics company and a mobility platform because an innovation changes the economics of autonomous fleets; a data-center operator and an energy-storage company because grid constraints alter the value of behind-the-meter resilience. Such edges must be evaluated for expert endorsement, not celebrated merely because they are novel.

6 Comparison with RAG, GraphRAG, Fine-Tuning, and Agent Memory

6.1 Traditional RAG

Traditional RAG retrieves relevant passages from an external index and conditions generation on them [3]. Its great advantage is that the non-parametric memory can be updated independently of the model. In the canonical interaction, however, the answer does not restructure the corpus. The system reads from memory but does not necessarily write analytical experience back into it.

Question-Induced Knowledge Architecture retains retrieval but adds a governed write path. The result of inquiry can become a new typed object with explicit lineage. The difference is not “retrieval versus no retrieval.” It is *read-only external memory versus recursively updated analytical memory*.

6.2 GraphRAG

GraphRAG addresses an important weakness of passage-level RAG: global sensemaking over a large corpus. It uses an LLM to derive an entity graph from source documents, detects communities, pregenerates community summaries, and uses those summaries to answer broad questions [1]. It therefore already demonstrates that LLM-derived graph structure can improve retrieval and synthesis.

The present architecture differs mainly in the causal role of inquiry and the status of the stored objects. In GraphRAG, graph construction is principally an indexing transformation derived from the corpus. In QIKA, a consequential question creates a reasoning episode whose output is itself a candidate addition to memory. Environmental shocks revise earlier episodes, and the revision history becomes part of the graph. GraphRAG asks: *How can graph structure improve answers over a corpus?* QIKA asks: *Can governed answers to successive decisions become reusable graph structure and thereby improve later decisions?*

6.3 Memory and reflection architectures

The boundary is not absolute. Several research traditions move beyond read-only RAG. Generative Agents store experience, retrieve it, and synthesize higher-level reflections for planning [7]. Reflexion stores linguistic feedback in episodic memory to improve later trials without updating model weights [9]. MemGPT gives an LLM control over hierarchical memory tiers [6]. HippoRAG combines LLMs, knowledge graphs, and personalized PageRank to integrate and retrieve new experiences [2]. Temporal graph memory explicitly models changing relations and periods of validity [8]. Self-Refine demonstrates iterative improvement through model-generated feedback [4].

These systems make it untenable to claim that persistent model-mediated memory is unique. The defensible contribution of QIKA is the integration of:

- purposeful decision questions as graph nodes and causal triggers;

- typed, domain-specific relations rather than generic memory items;
- shocks that revise earlier hypotheses instead of merely appending new observations;
- a protected source layer and a separate governed derived layer;
- explicit epistemic classes and human authorization gates; and
- evaluation of cumulative institutional decision quality across successive knowledge states.

Table 1: Architectural comparison. Categories describe common forms, not every possible implementation. Advanced variants can cross these boundaries.

Approach	Does prior reasoning persist?	Main distinction from QIKA
Fine-tuning	Yes, implicitly in weights	Expensive and opaque; weak episode-level provenance; proprietary facts may be absorbed into parameters.
Traditional RAG	Usually not as structured knowledge	External memory is commonly read-only during the analytical episode.
GraphRAG	Corpus-derived structure persists	Graph is mainly an index over source content, not necessarily a history of decision-conditioned revisions.
Self-refinement	Often only within a task	Improves an output but need not create a durable institutional graph.
Agent memory	Yes	Often optimized for agent continuity rather than governed domain knowledge and human decision accountability.
QIKA	Yes, after governance	Inquiry itself induces typed structure; later shocks revise it with explicit lineage.

7 The 100-Company M&A Laboratory

7.1 Objective and original state

The companion notebook operationalizes the theory as a transparent synthetic experiment. It creates an Obsidian vault containing 100 companies across ten sectors: banking, insurance, FinTech, energy, industrials, healthcare, consumer, telecom, software, and real estate. Each company has a region, capability, investment style, revenue growth, EBITDA margin, leverage, liquidity, interest sensitivity, FX exposure, innovation readiness, and valuation multiple.

The original state is deliberately sparse. It contains 100 factual company nodes and no asserted M&A links. This is crucial: the experiment does not pretend that relationships already exist and merely need to be displayed. The first relational layer must be caused by an analytical question.

With $n = 100$, the system evaluates

$$\binom{100}{2} = 4,950$$

unordered company pairs at each relevant ranking stage. The scoring functions are transparent and reproducible, while an optional API-backed LLM layer can generate richer narratives. This separation permits structural testing without making reproducibility depend on a proprietary model response.

7.2 Round 1: Q001 creates initial M&A hypotheses

The first question is:

Which pairs of companies are the strongest candidates for an M&A transaction?

The criteria are strategic compatibility, capability complementarity, geographic synergy, financial resilience, and growth potential. A simplified score for pair (i, j) is:

$$S_{ij}^{(1)} = 0.30C_{ij} + 0.20K_{ij} + 0.12G_{ij} + 0.18F_{ij} + 0.20R_{ij},$$

where C is sector compatibility, K capability complementarity, G geographic synergy, F financial resilience, and R growth potential. The system ranks all 4,950 pairs and writes only the top 30 as `INITIAL_MA_CANDIDATE` edges. Each edge carries Q001, episode E001, its score components, and the condition `PRE_SHOCK_BASELINE`.

The conceptual point is more important than the exact weights. The first question has transformed disconnected facts into a provisional decision map. Those edges are neither raw facts nor executed transactions. They are question-conditioned hypotheses.

7.3 Round 2: a macro shock creates exposure and revision links

The notebook adds a macroeconomic shock:

Policy rates rise by 250 basis points, local currencies depreciate by 18 percent, and aggregate demand contracts by 6 percent.

This event becomes node `S_MACRO_001`. Q002 asks which companies are most vulnerable and how the Q001 merger map should change. Company vulnerability is modeled as:

$$V_i = 0.25I_i + 0.22X_i + 0.25L_i + 0.18(1 - Q_i) + 0.10(1 - B_i),$$

where I is interest sensitivity, X FX exposure, L normalized leverage, Q liquidity strength, and B the growth buffer. The 25 most vulnerable firms receive `VULNERABLE_UNDER` links from the macro-shock node.

The architecture then revisits every initial merger hypothesis. A pair may be repriced, weakened, strengthened, or invalidated. Importantly, the original edge is not overwritten. The revision is linked to Q002, episode E002, the shock, and its parent experience E001. The graph now contains not only a fact about the environment but also a record of how that fact changed a prior decision.

7.4 Round 3: an innovation shock creates new possibilities

The second event is not merely adverse. It is an innovation shock:

Rapid adoption of agentic AI lowers integration costs and raises the value of proprietary data, distribution, cloud, cybersecurity, and automation assets.

Q003 asks which new M&A opportunities are created by this event. The system scores all 4,950 pairs again using innovation readiness, data-to-distribution complementarity, AI enablement, integration feasibility, and growth optionality. It excludes the original top pairs when selecting 30 newly emergent opportunities. These receive `INNOVATION_COMPLEMENT` edges linked to `S_INNOV_001`.

This stage illustrates why the graph cannot be reduced to static similarity. The innovation shock changes the economic meaning of existing attributes. Two companies that were not especially attractive under Q001 may become a high-value combination because a new technology changes integration costs or the value of their complementary assets.

7.5 Round 4: the final question reasons over accumulated experience

The final question is:

Which companies should now merge when we jointly require strategic compatibility, lower combined financial risk, and superior growth opportunities?

The system does not merely retrieve the original profiles. It has access to the initial ranking, macro vulnerability assessments, revisions, and innovation-created opportunities. A final illustrative score is:

$$S_{ij}^{(4)} = 0.34C_{ij}^* + 0.28F_{ij}^* + 0.23R_{ij}^* + 0.15N_{ij},$$

where C^* is integrated compatibility, F^* post-shock financial resilience, R^* growth opportunity, and N innovation optionality. The top 20 pairs receive `FINAL_MA_RECOMMENDATION` edges with full lineage across Q001–Q004.

7.6 What the visualization does and does not prove

The before-and-after graph is visually striking, but density is not knowledge. A random linker could create a denser graph. The visualization establishes only that the architecture can externalize a sequence of question- and shock-conditioned relations. Scientific validation requires a separate test: whether experts endorse those links, whether they improve held-out decisions, whether their provenance is intact, and whether the system resists self-reinforcing error.

This distinction is essential to the paper’s credibility. The notebook is a computational laboratory for the hypothesis, not proof of the hypothesis.

The Vault Changes When Questions Become Persistent Experience

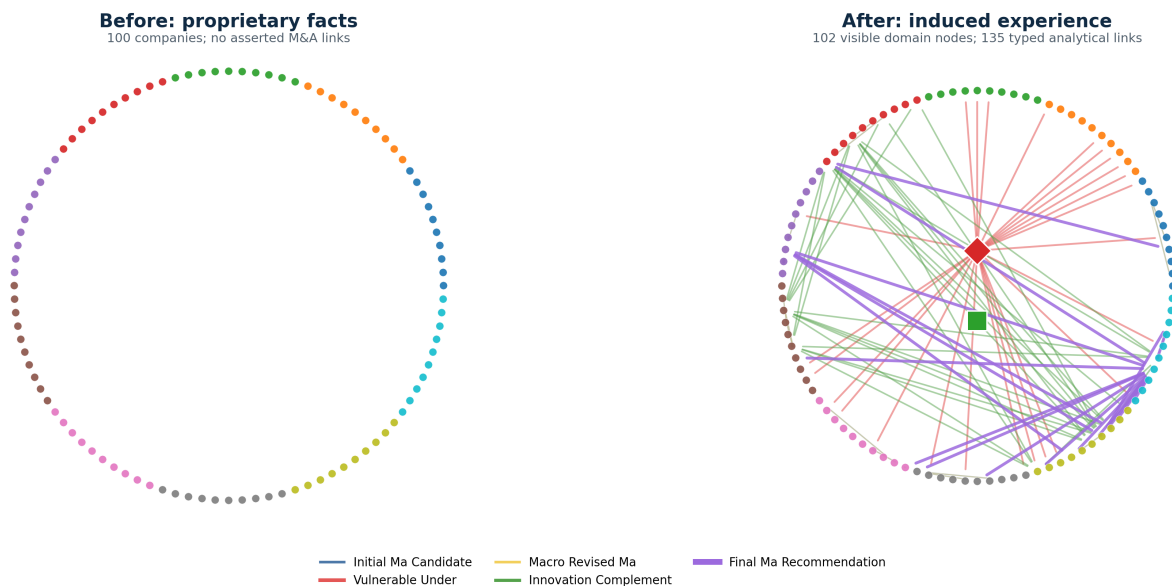


Figure 2: Before and after views generated from the notebook’s deterministic 100-company architecture. Company positions remain fixed. The enriched graph contains initial M&A hypotheses, macro vulnerability and revision links, innovation-created opportunities, and final recommendations. Question and reasoning-episode nodes are omitted from this domain view for legibility.

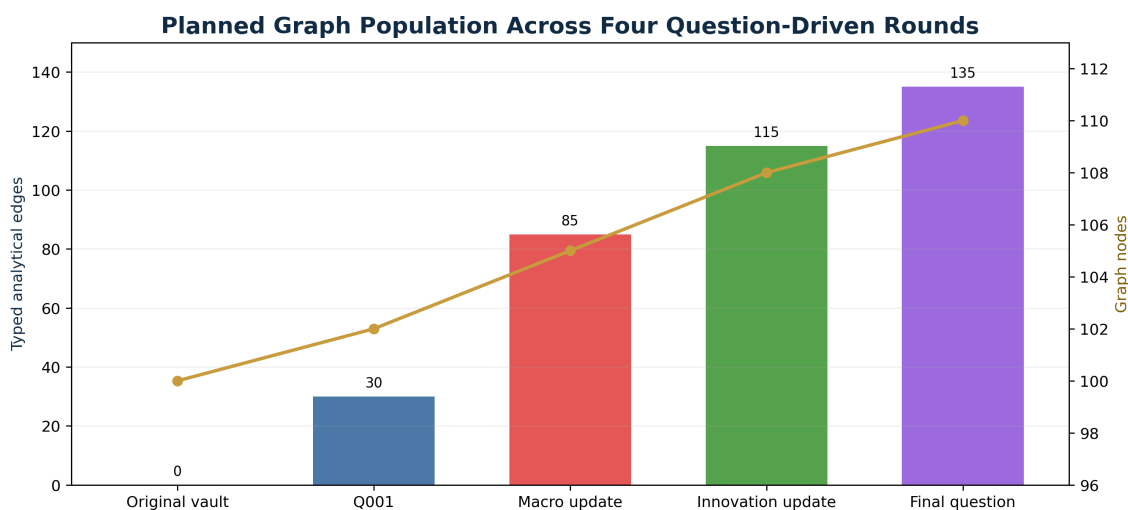


Figure 3: Planned accumulation of nodes and typed analytical edges. Counts describe the notebook’s designed outputs, not evidence that the induced knowledge has already passed expert validation.

8 A Stronger Experimental Design

8.1 Comparative arms

To isolate the value of recursive enrichment, the experiment should compare at least five systems under equal model, token, and evidence budgets:

1. **Raw-facts baseline:** the model receives selected company facts but no retrieval index or stored reasoning.
2. **Static RAG:** retrieval over the original company vault; answers are not written back.
3. **Static GraphRAG:** an entity graph and community summaries are built from the original source corpus, but decision outputs do not become memory.
4. **Question-induced, ungoverned memory:** reasoning outputs are stored automatically.
5. **Question-induced, governed memory:** only validated, typed, and provenance-complete objects enter the derived layer.

The fourth arm is necessary. Without it, one cannot determine whether gains come from persistence itself or from the governance that protects persistence from error compounding.

8.2 Held-out tasks

Construction questions must be separated from evaluation questions. After the four enrichment rounds, the system should face unseen evidence chunks and new decision tasks, for example:

- a regulatory restriction that changes permissible ownership;
- a commodity-price shock with sector-specific transmission;
- an unexpected deterioration in one recommended firm’s liquidity;
- a cross-border antitrust constraint;
- a new entrant whose attributes overlap several prior cases; and
- an ambiguous scenario requiring explicit uncertainty rather than a forced recommendation.

The critical test is whether the enriched system makes better use of prior experience without treating prior model outputs as incontrovertible evidence.

8.3 Metrics

The measurement framework should include:

Decision utility. Expert ranking quality, pairwise preference accuracy, and the economic coherence of recommendations.

Factual fidelity. Whether every factual component is supported by the protected source corpus or an identified external source.

Relational precision. The proportion of induced edges endorsed by domain experts.

Distant-edge value. Expert endorsement of useful relations between objects that are not close in embedding space.

Revision responsiveness. Whether shocks appropriately strengthen, weaken, invalidate, or create links.

Calibration. Alignment between confidence and expert correctness.

Provenance integrity. Completeness of question, evidence, model, condition, timestamp, and parent-revision lineage.

Counterfactual discipline. Whether conclusions change when the governing shock or assumption is removed.

Knowledge contribution. The marginal performance gain attributable to retrieved induced experience, measured through ablation.

Epistemic pollution. The rate at which unsupported generated claims contaminate later answers.

A useful contribution measure for stored object k is:

$$KC(k) = \mathcal{P}(V_t) - \mathcal{P}(V_t \setminus \{k\}),$$

estimated over held-out tasks. This makes it possible to prune objects that increase density but do not improve reasoning.

8.4 Robustness tests

The architecture should be tested across different model families, retrieval strategies, random seeds, and shock orders. A genuine knowledge architecture should not depend entirely on the prose style or idiosyncratic priors of one model. Particularly revealing tests include:

- swapping the generating model while keeping the vault fixed;
- using one model to generate and a different model to critique;
- reversing the order of macro and innovation shocks;
- withholding the original reasoning narrative while retaining typed edges;
- withholding edges while retaining narratives;
- introducing deliberately false or adversarial evidence; and
- measuring whether expert disagreement is preserved rather than collapsed into a single artificial consensus.

9 Epistemic Architecture and Governance

9.1 The protected source layer and the derived layer

The original vault should remain an identifiable research corpus. Model outputs should not silently enter the same epistemic category as source documents. A robust implementation separates:

1. **Source facts:** imported or human-authored records with stable provenance.
2. **Model hypotheses:** proposed relations or explanations.
3. **Shock assessments:** conditional analyses tied to specified events.
4. **Recommendations:** decision proposals requiring authorization.
5. **Human decisions:** approved actions and the identity of the approving authority.
6. **Outcomes:** later observations that test the recommendation.

The derived graph can point to protected source notes but should not rewrite them. Promotion from candidate to approved knowledge should be a logged state transition, not a copy-and-paste operation that erases origin.

9.2 Minimum metadata for every induced relation

Every stored edge should contain:

- source and target identifiers;
- typed relation;
- question and reasoning-episode identifiers;
- source evidence references;
- model and prompt version;
- score and confidence;
- assumptions and governing condition;
- epistemic class and approval status;
- valid-from and valid-to timestamps;
- parent relation, if revised;
- superseding relation, if invalidated; and
- human reviewer and decision, where applicable.

This metadata is not bureaucratic decoration. It is what makes recursive memory scientifically and institutionally safe. Without it, the system cannot distinguish original evidence from its own earlier interpretation.

9.3 The authorization ladder

A practical write-back policy can use the following states:

generated $\rightarrow$ quarantined $\rightarrow$ validated $\rightarrow$ approved $\rightarrow$ active $\rightarrow$ superseded.

Low-risk descriptive links may be approved automatically after deterministic checks. High-impact recommendations, especially in finance, law, tax, or regulation, should require human authorization. No graph edge should itself execute a transaction. The knowledge system prepares decision intelligence; it does not erase accountability.

10 Why This Is a Different Kind of Training

The word “training” must be used carefully. Conventional training changes parameters by optimizing a loss function over examples. QIKA does not require that. It changes the structured environment against which future inference is performed. Three properties justify describing it as an externalized learning process:

1. **Persistence:** selected experience survives beyond the context window.
2. **Behavioral consequence:** later outputs change because that experience is retrieved.
3. **Revision:** new evidence can update the validity and meaning of earlier knowledge objects.

The process resembles case-based learning or organizational memory more than weight optimization. An investment bank does not retrain its employees’ brains when it writes a transaction post-mortem. Yet the institution can learn if the post-mortem is retrievable, trusted, connected to later cases, and incorporated into decision routines. The vault plays the role of externalized institutional memory; the LLM helps convert distributed records into usable symbolic structure.

This also changes the meaning of personalization. A model need not absorb proprietary material into opaque parameters to become useful in a specialized domain. Instead, it can repeatedly operate against a governed, organization- specific memory that remains inspectable, portable, and revocable. The institution owns the evolving knowledge substrate even if the underlying model is replaced.

11 Implications for Richer Bodies of Knowledge

11.1 From archives to analytical organisms

Most organizational repositories are accumulations. They grow by adding files, not by explicitly representing how one analysis changes another. QIKA suggests a transition from archival growth to relational growth. A mature vault would contain not only “what we knew” but:

- what we asked;

- why the question mattered;
- what alternatives were considered;
- what we concluded under which assumptions;
- what shock changed the conclusion;
- who approved the revised view; and
- what outcome later confirmed or contradicted it.

Such a vault is not omniscient. It is more valuable precisely because it can represent uncertainty, disagreement, and supersession.

11.2 Institutional applications

The M&A case is one example. The same architecture can support:

Tax planning. Regulations, entity structures, advice, quarterly code changes, and the lineage of revised cross-border recommendations.

Legal strategy. Facts, precedents, arguments, counterarguments, rulings, and the conditions under which a theory of the case changed.

Scientific research. Papers, experiments, hypotheses, failed replications, and new questions induced by conflicting results.

Financial supervision. Institutions, exposures, macro scenarios, policy interventions, and the transmission paths revealed by successive stress tests.

Corporate strategy. Competitors, technologies, capabilities, scenarios, decisions, and post-decision outcomes.

Education. Concepts, student questions, misconceptions, explanations, and the pedagogical paths that proved effective.

In each domain, the highest-value content may be the relation between events and revised judgments, not the events alone.

11.3 Model portability and institutional continuity

Because the learning resides in the governed external layer, the institution can change models without losing its analytical history. This is strategically important. Model vendors, prices, context windows, and capabilities will change. A portable graph of questions, evidence, hypotheses, revisions, and decisions preserves institutional continuity. The model becomes a replaceable reasoning engine operating over a durable knowledge architecture.

12 Risks, Failure Modes, and Limits

The central danger is epistemic self-consumption. If generated hypotheses are retrieved later as though they were independent facts, the system can manufacture apparent corroboration from its own prior outputs. Persistence amplifies both insight and error.

12.1 Error compounding and synthetic consensus

A plausible but false relation can become highly connected, frequently retrieved, and increasingly influential. Later answers may repeat it, producing the illusion that multiple analyses agree when they all descend from one unsupported episode. Lineage-aware retrieval must therefore discount non-independent descendants and expose common ancestry.

12.2 Path dependence

Early questions shape graph topology. What the organization chooses to ask determines which relations become visible and retrievable. A vault repeatedly interrogated for merger opportunities may underrepresent labor, environmental, or operational risks. Question diversity is therefore a governance variable. The system should preserve an agenda of unanswered and adversarial questions, not only a memory of accepted conclusions.

12.3 Topology-induced bias

Once edges affect retrieval, well-connected nodes become easier to retrieve, which can make them even more connected. This preferential-attachment dynamic may crowd out weak signals and novel entrants. Evaluation should compare graph-based retrieval with source-level sampling and include deliberate exploration of low-degree nodes.

12.4 Fluency, authority, and model priors

The model may bind proprietary facts to a familiar but inappropriate public schema. Fluent explanations can encourage reviewers to accept weak relations. The remedy is not merely more confidence scores. It requires evidence-level review, alternative hypotheses, model diversity, and explicit tests of counterfactual sensitivity.

12.5 Temporal validity

Recommendations are especially perishable. A relationship valid before a rate shock may be invalid afterward; a technology opportunity may disappear when regulation changes. Graph objects require time intervals and supersession semantics. Deletion destroys learning; timeless accumulation destroys relevance.

12.6 Confidentiality and control

The architecture may expose proprietary data to external model services. Deployment must address data minimization, encryption, access control, regional requirements, retention policies, model-provider terms, and redaction. Governance of knowledge formation cannot be separated from governance of data movement.

12.7 Limits of the current notebook

The companion laboratory is synthetic and uses transparent scoring formulas. It demonstrates architecture, provenance, and graph population; it does not establish superior real-world M&A performance. Its optional LLM layer adds narrative interpretation but does not by itself constitute an expert validation study. Real evidence would require blinded domain review, held-out historical cases, transaction outcomes, robustness across models, and careful control for information leakage.

13 Research Propositions

The theory yields seven falsifiable propositions.

H1: Semantic densification. Purposeful questions produce a greater number of expert-endorsed, typed relations than source-only extraction.

H2: Beyond similarity. A non-trivial share of useful induced edges connect items that are distant in embedding space.

H3: Contextual compounding. Governed induced experience improves performance on held-out decision tasks relative to static RAG and static GraphRAG.

H4: Revision value. Explicit shock-conditioned revisions improve adaptation compared with appending shock documents without revising prior hypotheses.

H5: Governance premium. Governed persistence outperforms ungoverned persistence after several rounds because it limits epistemic pollution.

H6: Portability. A new model using the same governed vault retains a material share of the accumulated performance advantage.

H7: Saturation. Marginal useful structure eventually declines, and indiscriminate graph densification reduces performance.

Failure would also be informative. If static GraphRAG performs equally well, the claimed value of question-conditioned experience would be limited. If ungoverned memory performs better, the governance design may be too costly or restrictive. If model replacement destroys the benefit, the knowledge layer may encode model-specific rhetoric rather than portable institutional structure. A serious experiment must allow these outcomes.

14 Conclusion: Generative AI as a Partner in Knowledge Formation

The most consequential use of generative AI may not be the production of one more answer. It may be the deliberate formation of a knowledge environment in which later answers become better, more connected, and more accountable.

An original vault provides the evidence substrate. A purposeful question creates a problem of relevance. The language model combines local evidence with broad parametric priors and proposes new relational structure. A shock tests that structure and creates a revision episode. Governance separates fact from inference, validates provenance, and decides what may persist. The next question is then posed to a different epistemic environment—one jointly produced by the proprietary corpus, prior model reasoning, external events, and human judgment.

This is why the architecture is more subtle than “adding context.” Context is no longer a static input. It becomes an evolving product of inquiry. Nor is the mechanism simply GraphRAG, although graphs and graph retrieval are central to it. The defining feature is recursive, question-conditioned, governed write-back: the conversion of analytical episodes into externalized institutional experience.

The proposition can be stated compactly:

The model need not change for the cognitive system to learn. When purposeful questions, evidence, conditional hypotheses, shocks, revisions, and decisions are preserved as governed graph objects, the model–vault system can accumulate a form of externalized experience. The result is not a larger prompt. It is a living, inspectable, and revisable body of knowledge.

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